

FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY GRADE 10: ACIDS AND BASES (SUB-STRAND 3.1)

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

In this Final Explanation you answer the driving question: Does the alkalinity of a substance determine its uses in day to day life? Why are patients with hyperacidity issues given antacid medications? Use the evidence you collected in Lessons 1 to 10 and your final particle model (H^+ (aq) and OH^- (aq) ions). Write in full sentences in the space under each prompt. In every answer, give a claim, evidence (use numbers and observations) and reasoning (use particles and ions). Use the pH data below. Refer to the Kisumu products, Mr Kamau in Kericho and the pharmacist in Nairobi where you can.

Data table (pH measured with a pH meter at 25 °C):

Substance	pH
Lemon juice (Kisumu market)	2.0
Vinegar (5% ethanoic acid)	2.8
Pure water	7.0
Baking soda solution (as used for mandazi)	8.3
Bar soap solution	10.0

Part 1: Sorting products that look alike

Lemon juice, vinegar, baking soda solution and soap solution look similar in clear cups, yet Otieno found lemon sour and Wanjiru found soap slippery. Using the pH table and the indicator results from your tests (litmus, red cabbage extract and universal indicator), classify each product as acid, base or neutral. Explain how the indicators and pH values agree even though the colours differ. Which product in each pair is more acidic or more basic: lemon juice and vinegar, soap and baking soda? Give the pH evidence.

Write your answer here:

Part 2: What is inside acids and alkalis?

Explain at the particle level why lemon juice is sour and has a low pH, and why soap feels slippery. Use the Arrhenius definitions of an acid and an alkali, and the concentration data: lemon juice $\text{H}^+ = 1 \times 10^{-2} \text{ mol/L}$, vinegar H^+ about $1.6 \times 10^{-3} \text{ mol/L}$, water $1 \times 10^{-7} \text{ mol/L}$, soap $\text{OH}^- = 1 \times 10^{-4} \text{ mol/L}$. Explain how much more H^+ lemon juice has than vinegar, and say how you know. Describe how the Model you drew in Lesson 1 changed.

Write your answer here:

Part 3: How acids and bases behave

When vinegar was added to baking soda it fizzed. Magnesium ribbon was placed in dilute hydrochloric acid. Describe what you observed and how you tested each gas. Explain why these reactions happen with acids but not with soap or water, and write word equations for the two reactions. Then explain what happens, in terms of ions, when an acid and a base are mixed.

Write your answer here:

Part 4: Strength, safety and uses

Otieno asked: if both are acids, why is lemon juice lower on the graph than vinegar? Use the data for 0.1 M hydrochloric acid (pH 1), 0.1 M ethanoic acid (pH 3), 0.1 M sodium hydroxide (pH 13) and 0.1 M ammonia solution (pH 11) to explain the difference between strength and concentration. Then explain why a weak base such as magnesium hydroxide is a safer antacid than a strong base, and why soap is useful for cleaning but harmful if swallowed.

Write your answer here:

Part 5: Answering the driving question

Write your final Claim-Evidence-Reasoning answer to the driving question: Does the alkalinity of a substance determine its uses in day to day life? Why are patients with hyperacidity issues given antacid medications? Use at least three uses (for example vinegar, soap, baking soda, Mr Kamau's lime in Kericho, the pharmacist in Nairobi). Include the stomach pH values and the soil pH values. Say what else, besides alkalinity, matters.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim answers the driving question	Gives a clear claim that alkalinity, with strength and safety, largely determines uses, and states why antacids are given for hyperacidity.	Gives a claim about uses and antacids but leaves out strength or safety.	Claim is missing, unclear or does not address both parts of the question.
Use of evidence and data	Uses exact pH values, concentrations, indicator colours, reaction tests and stomach/soil pH changes correctly in every part.	Uses most data correctly, with one or two missing or small errors.	Uses little data, or data is incorrect or not linked to the claim.
Particle-level reasoning	Explains using H^+ (aq) and OH^- (aq), the Arrhenius definitions, the 10-fold pH change and H^+ (aq) + OH^- (aq) gives H_2O (l) accurately.	Mentions H^+ and OH^- ions correctly but explanations are incomplete.	Does not use ions, or confuses acids and bases.
Strength, concentration and safety	Clearly separates strength from concentration using the HCl, ethanoic acid, NaOH and ammonia data, and links this to safe antacid and soap use.	Describes strength correctly but does not link to safety clearly.	Confuses strength with concentration or omits safety.
Kenyan context and communication	Links to Kisumu products, Kericho lime and the Nairobi pharmacist; uses correct terms in clear, organised sentences.	Uses some contexts and mostly correct terms.	Little context; terms are misused or writing is hard to follow.

ATTRIBUTION AND LICENSING

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